

Slide 1 – Introduction (Hook Story)

Today we are not learning cybersecurity as theory — we are learning how small daily mistakes can create big security incidents.

A developer uploaded code to GitHub with .env file included containing AWS key, DB password, API token. Within 20 minutes attacker accessed cloud account and created multiple servers. Company suffered financial loss.

Lesson: Attackers scan internet continuously for mistakes.

Slide 2 – Security-First Mindset

Security should be part of daily development habit, not last step. Example: attacker sends malicious script as input and application stores it, causing XSS attack.

Always think: What if attacker sends unexpected input?

Slide 3 – Why Security Still Fails

Companies install many tools but ignore alerts. One unusual login alert ignored for days led to database breach.

Tools alone do not provide security. People must actively monitor.

Slide 4 – Modern Attack Surface

Attackers target GitHub repos, APIs, CI/CD pipelines, npm packages and developer laptops.

Example: malicious npm package steals API keys.

Slide 5 – Common Developer Mistakes

Hardcoded passwords in code expose database access.

Always store secrets in environment variables and never commit .env files.

Slide 6 – Credential Security Risks

Using same password across platforms allows chain attacks.

Use password manager, MFA and rotate tokens regularly.

Slide 7 – API Security Awareness

Changing user ID in API request exposes other users' data if authorization is missing.

Always validate authentication and authorization on backend.

Slide 8 – Secure Coding Practices

SQL Injection occurs when input is not validated.

Use parameterized queries and validate user input.

Slide 9 – Cloud Misconfiguration

Public cloud storage buckets expose confidential documents.

Always check IAM permissions and restrict access.

Slide 10 – Social Engineering

Fake GitHub login email steals credentials.

Always verify domain before entering login details.

Slide 11 – Developer Machine Security

Cracked software may contain malware that steals SSH keys.

Secure laptop using encryption and updated OS.

Slide 12 – Logging and Zero Trust

Without monitoring, attacker activity may go unnoticed.

Zero Trust means always verify user identity and permissions.

Conclusion

Attackers need only one mistake. Always validate inputs, secure credentials, enable logging and follow least privilege.